

Practical Portfolio Operational Plan: Supporting Portfolio (المحفظة الداعمة)

1. Executive Overview & Grading Rubric Alignment (نظرة عامة ومواءمة التقييم)

This operational plan defines the requirements for the "Supporting Portfolio" (المحفظة الداعمة), a mandatory component of the BAXZ1313 "Varsity Cyber Clash" project. This document serves as the primary instrument for the lab instructor to evaluate your team's technical implementation skills.

Grading and Assessment (التقييم والدرجات):

- **Proof of Competency:** Successful completion of the four stations documented herein is required to secure the 20 marks allocated to "Proof of competency skills" (Item 4) in the Project Rubric.
- **Strict Evidence Requirement:** Incomplete evidence, blurry screenshots, or missing documentation for any station will result in an automatic score of 0–1 for that evaluation element.
- **Complexity Justification:** To secure the 5 marks for "Complexity of the Project" (Item 6), students must provide technical justifications for their chosen addressing schemes and hardware configurations within this portfolio.

Objective: To demonstrate professional-grade proficiency in physical media construction, logical addressing, wireless deployment, and resource management.

2. Station 1: Physical Layer - Media Construction (المحطة الأولى : (الطبقة الفيزيائية - بناء الوسائط

This station validates the team's ability to construct the physical backbone of the "Varsity Cyber Clash" network using CAT6 cabling for stationary gaming workstations.

Technical Instructions & Quality Standards

1. **Preparation:** Strip 1 inch of the outer jacket. Untwist and straighten pairs.
2. **Wiring Standard:** Arrange wires strictly according to the T568B standard (Table below).
3. **Technical Checkpoint (Instructor Priority):** Trim wires to 0.5 inches. Ensure the cable jacket is positioned inside the RJ45 connector body before crimping. Cables with exposed internal wires outside the connector will be rejected for lack of strain relief.
4. **Termination:** Use a professional crimping tool to secure the RJ45 plug.
5. **Verification:** Test the cable using a LAN Tester to ensure 1-to-1 continuity.

T568B Wiring Standard

Pin	Wire Color
1	White-Orange

2	Orange
3	White-Green
4	Blue
5	White-Blue
6	Green
7	White-Brown
8	Brown

Evidence Requirements

- **Photograph 1 (Pre-Crimp):** A high-resolution close-up of the wire alignment inside the RJ45 connector *before* crimping. The order must be clearly visible.
- **Photograph 2 (Validation):** A photo of the LAN tester showing the sequential LED pattern (1 through 8) on both units.

3. Station 2: Logical Configuration - IPv4 & VLSM (المحطة الثانية : (التكوين المنطقي - العنونة والتقسيم)

Students must configure a workstation to reside within the tournament's private IP block (10.0.0.0/8). While the lab uses a flat /8 for demonstration, teams must justify their sub-range selection as part of a broader VLSM strategy for the 200+ tournament participants.

Configuration Procedure (Windows OS)

1. Navigate to Control Panel > Network and Sharing Center > Change adapter settings.
2. Right-click the Ethernet interface > Properties.
3. Select Internet Protocol Version 4 (TCP/IPv4) > Properties.
4. Select Use the following IP address and input:
 - IP Address: 10.0.0.10 (Ensure no conflict with Station 3 DHCP range).
 - Subnet Mask: 255.0.0.0
 - Default Gateway: 10.0.0.1

Evidence Requirements

- Screenshot 1: Full-screen capture of the "TCP/IPv4 Properties" dialog box showing the manual IP and Mask.
- Screenshot 2: Command Prompt window executing ipconfig /all. The capture must show the system clock and the physical address (MAC) of the adapter.

4. Station 3: Wireless Infrastructure Setup (المحطة الثالثة: إعداد البنية اللاسلكية)

This station simulates providing connectivity for the "Varsity_Spectators" group. The setup must ensure isolation from the gaming subnets while remaining within the tournament's overall 10.0.0.0/8 allocation.

Mandatory Configuration Parameters

- **SSID: Varsity_Spectators**
- **Security Type: WPA2-Personal (AES Encryption)**
- **DHCP Server: Enabled**
- **IP Range: 10.0.5.100 to 10.0.5.200 (Subnet Mask: 255.0.0.0)**

Implementation Requirements

Access the Access Point (AP) web interface. Configure the SSID and Security menus first. In the LAN/DHCP settings, define the specific range mentioned above to demonstrate the application of the VLSM sub-range for non-critical users (spectators).

Evidence Requirements

- **Screenshot 1: The AP management interface showing the Wireless Settings (SSID and Security).**
- **Screenshot 2: The DHCP Server settings page showing the specific start/end IP range assigned to spectators.**

5. Station 4: Connectivity Verification & Resource Sharing (المحطة الرابعة: التحقق من الاتصال ومشاركة الموارد)

This station confirms the network is ready for the "Tournament Management System." You must demonstrate the ability to share critical "Game Files" or "Tournament Databases" between nodes using advanced permissions.

Task A: End-to-End Connectivity

Execute the ping command from the configured workstation to another node (e.g., the Gateway or a second PC).

Task B: Advanced Resource Sharing (Tournament File Server)

Do not use the "Simple Sharing" shortcut. Use Advanced Sharing to simulate a secure environment:

1. Right-click the "Tournament_Data" folder > Properties > Sharing tab.
2. Click Advanced Sharing > Check "Share this folder."
3. Click Permissions and ensure "Everyone" is restricted to Read, while "Admins" have Full Control.
4. Access the folder from a remote PC using the UNC path:
\\10.0.0.10\\Tournament_Data.

Evidence Requirements

- Screenshot 1: Command Prompt showing a successful Ping response (0% loss) with the target IP visible.
- Screenshot 2: Remote PC view showing the "Tournament_Data" directory being successfully accessed via the network path.

6. Final Submission Checklist (قائمة التحقق النهائية للتسليم)

Station (المحطة)	Required Evidence (الأدلة المطلوبة)	Verified (تم التحقق)
1: Physical Layer	Photos of T568B alignment (Pre-crimp) & LAN Tester results	[]

2: Logical Config	TCP/IPv4 Properties screenshot & ipconfig /all	[]
3: Wireless Setup	Screenshots of AP SSID, WPA2 Security, & DHCP Range	[]
4: Connectivity	Ping response capture & Shared Folder access (UNC Path)	[]

Final Directive: All evidence must be compiled into a single PDF document. Per Section 1.0.2 of the Project Specification, submission is due via ULearn by Sunday noon, 14 June 2026.

Penalties for Late Submission:

- **1–3 Days Late: 25% mark deduction.**
- **> 3 Days Late: 50% mark deduction.**